

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2020-2021

**SYSTEMS AND ARCHITECTURE
(COMP 1030)**

Time allowed ONE hour

An additional 30 minutes to cover IT issues and upload time

Answer TWO questions out of THREE

Submit your answers containing all the work you wish to have marked as a **single PDF file, with each page in the correct orientation, to the appropriate dropbox on the module's Moodle page.**

Use the standard naming convention for your document: [Student ID]_[Module Code]. Write your student ID number at the top of each page of your answers. Do not include your name.

Although you may use any notes or resources you wish to help you complete this open-book examination, the academic misconduct policy that applies to your coursework also applies here. You must be careful to avoid plagiarism, collusion or false authorship. This statement refers to, and does not replace, the University policy which stipulates severe penalties for academic misconduct.

Staff are not permitted to answer assessment or teaching queries during the period in which your open-book examination is live. If you spot what you think may be an error on the exam paper, note this in your submission but answer the question as written.

IMPORTANT: Lateness penalties will not be applied, and late submission will not be accepted.

1. Networks (25 Marks)

- (a) With the help of diagrams, briefly describe the operations of the three types of fiber-optic cables currently available. Of the three, which is the most efficient? Why?
[5 Marks]
- (b) (i) Describe the operation of following non-persistent, 1-persistent and p-persistent CSMA/CD protocols.
[3 Marks]
- (ii) Which persistent algorithm would you like to recommend for LAN data transmission? Why?
[3 Marks]
- (c) Distinguish between circuit switching method and virtual circuit method. What happens in virtual circuit packet switched subnet if a node or communications link along the virtual path fails?
[6 Marks]
- (d) Describe *isolated routing* and explain how it may gather the wrong or insufficient data.
[4 Marks]
- (e) What is Collision Avoidance and how is that achieved in CSMA/CA protocol?
[4 Marks]

2. Binary arithmetic, Processor Registers and Addressing modes (25 Marks)

- (a) Based on the register and memory contents specified, find the value of the operand for each of the following addressing mode.

- (i) Immediate Addressing Mode
- (ii) Register Addressing Mode
- (iii) Register Indirect Addressing Mode
- (iv) Relative Addressing Mode
- (v) Indexed Register Addressing Mode
- (vi) Indirect addressing mode [3 Marks]

Instruction format:

M= 1	Opcode	Address = 0010
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Memory

Address	Data
0000	0010
0001	0011
0010	1111
0011	1000
0100	0001
0101	0011
0110	1000
0100	0100
0101	0000
0111	0010

Program Counter = 0000

Index Register = 0001

R1 = 0010

- (b) Based on the register and memory contents specified, find the value of the register R0 for each of the following operations.

Memory

Address	Data
0000	1111
0001	1100
0010	1110
0011	0100
0100	0001
0101	0011
0110	1000
0100	0100
0101	0000
0111	0010
1111	1111

R1 = 1111

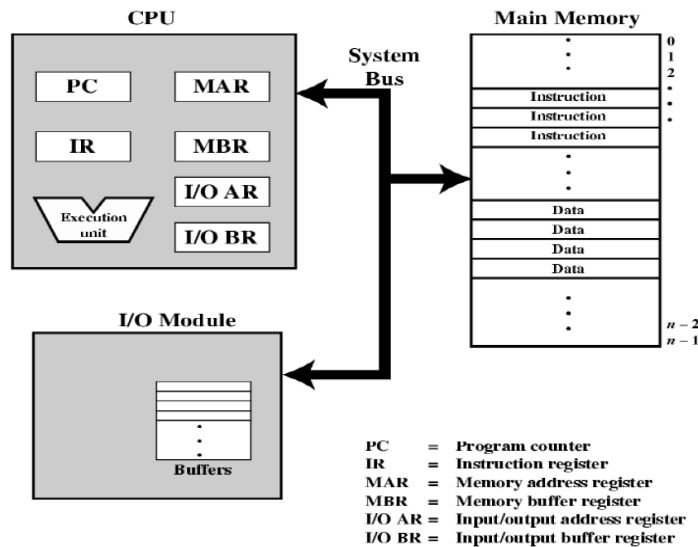
R2 = 0000

R0 = ?

- (i) MOV R0, #1111
- (ii) LDR R0, R1
- (iii) LDR R0, [R2]
- (iv) LDR R0, [R1,R2]

[2 Marks]

- (c) Suppose that the programming instructions are found at the locations in main memory as shown in the following diagram:



Briefly explain how these programming instructions are executed by the CPU using the PC, IR, MAR, MBR and Execution unit.

[6 Marks]

- (d) Represent the number **01110010 . 0110**₂ by using the given single and double precision formats given below.

Single precision format

1	8 bits	23 bits
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Sign bit Exponent in excess 127 Mantissa fraction

Double precision format

1	11 bits	52 bits
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Sign bit Exponent in excess 1023 Mantissa fraction

[4 Marks]

- (e) Divide the decimal number 12 by 2 in binary using a binary division algorithm. Use the register B to store the dividend and the Y register to store the divisor and A for the accumulator. Show the contents of registers B, Accumulator and Y register during each stage of the division process.

[8 Marks]

- (f) What is the difference between the direct and indirect addressing technique? How many references to memory are needed for each form of address mode to bring an operand into a processor register?

[2 Marks]

3. Interrupts, I/O and Instruction Pipelining (25 Marks)

- (a) Interrupt-driven I/O, though more efficient than simple programmed I/O, still requires the active intervention of the processor to transfer data between memory and an I/O module, and any data transfer must traverse a path through the processor. Thus, both these forms of I/O suffer from two inherent drawbacks.

Briefly describe the two drawbacks.

[4 Marks]

- (b) (i) What is a data hazard? Name and describe a hardware technique that overcomes most data hazards.

[4 Marks]

- (ii) Identify any four types of register data dependencies in the following piece of code.

L1: MUL r2, r3, r4

L2: ADD r3, r2, r4

L3: ADD r4, r2, r4

L4: SUB r3, r4, r3

[4 Marks]

- (c) When a multiple device interrupts have occurred; how does the processor decide which one to process?

[8 Marks]

- (d) When DMA module takes control of a bus, and while it remains control of the bus, what does the processor do?

[3 Marks]

- (e) What, if any, is the difference between a multicore processor and a coprocessor?

[2 Marks]